

# UChicago REU Notes Week 5

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## Contents

### 1 Categorical Preliminaries

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To be honest, last week's note isn't as successful, so we'll mainly focus on the category theory from now on. Here, we assume that  $G$  is a finite group, and the categories (mentioned as objects of  $\mathbf{Cat}$  or similar objects) are small.

## 1 Categorical Preliminaries

This part is devoted to understand some basics. I'll fill in the details for preliminaries on  $G$ -categories later, together with adjunction  $(\mathbf{Ob} \dashv \mathcal{E})$  between  $\mathbf{Cat}$  and  $\mathbf{Set}$ .

Given a category  $\mathcal{A}$ , the functor category  $\mathbf{Cat}(\mathcal{E}(G), \mathcal{A})$  is automatically a  $G$ -category, as each element  $g \in G$  can be realized as an automorphism functor on the indiscrete category  $\mathcal{E}G$ , so it admits both a left/right action by precomposing either  $g^{-1}$  or  $g$  as automorphism of  $\mathcal{E}G$ .

Similarly, if  $\mathcal{A}$  itself already has a left/right  $G$ -action, then it defines a left/right  $G$ -action on  $\mathbf{Cat}(\mathcal{B}, \mathcal{A})$  by postcomposition; similarly if  $\mathcal{B}$  is a  $G$ -category, then precomposition also defines one. Combining the two, it admits a conjugation action on  $\mathbf{Cat}(\mathcal{B}, \mathcal{A})$  if both  $\mathcal{B}, \mathcal{A}$  admits a left action.

Given  $G$ -categories  $\mathcal{A}, \mathcal{B}$ , denote  $\mathbf{Cat}_G(\mathcal{B}, \mathcal{A}) := \mathbf{Cat}(\mathcal{B}, \mathcal{A})$ , and  $G\mathbf{Cat}(\mathcal{B}, \mathcal{A}) := \mathbf{Cat}(\mathcal{B}, \mathcal{A})^G$  as the  $G$ -fixpoint (i.e. all functors commuting with  $G$ -actions).

**Proposition 1.1.** *There is a  $G$ -equivariant isomorphism when giving  $\mathcal{A}, \mathcal{B}, \mathcal{C}$  three  $G$ -categories:*

$$\Phi : \mathbf{Cat}_G(\mathcal{A} \times \mathcal{B}, \mathcal{C}) \cong \mathbf{Cat}_G(\mathcal{A}, \mathbf{Cat}(\mathcal{B}, \mathcal{C}))$$

*Proof.* We recognize this as a non-equivariant statement first, then show if all three are  $G$ -categories (with  $G$  acts diagonally on  $\mathcal{A} \times \mathcal{B}$ ), such isomorphism is  $G$ -equivariant.

### I. Construction of Functor:

Given a functor  $F : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ , for any object  $X \in \mathcal{A}$  (together with identity  $\mathrm{id}_X : X \rightarrow X$ ), define a functor  $F_X : \mathcal{B} \rightarrow \mathcal{C}$  by  $F_X(Y) := F(X, Y)$ , and given any morphism  $g : Y \rightarrow Y'$  in  $\mathcal{B}$ , define the morphism  $F_X g : F_X(Y) \rightarrow F_X(Y')$  as the morphism  $F(\mathrm{id}_X, g) : F(X, Y) \rightarrow F(X, Y')$ . The routine check that this is functorial, using the product category definition:

- Given  $\mathrm{id}_Y : Y \rightarrow Y$ , we have  $F_X \mathrm{id}_Y = F(\mathrm{id}_X, \mathrm{id}_Y) = \mathrm{id}_{F(X, Y)} = \mathrm{id}_{F_X(Y)}$ .

- Given  $g : Y \rightarrow Y'$ ,  $g' : Y' \rightarrow Y''$ , we have

$$F_X(g' \circ g) = F(\text{id}_X, g' \circ g) = F(\text{id}_X, g') \circ F(\text{id}_X, g) = F_X g' \circ F_X g$$

Now, given a morphism  $f : X \rightarrow X'$  in  $\mathcal{A}$ , we shall define a natural transformation  $Nf : F_X \rightarrow F_{X'}$ . Realistically, for any  $Y \in \mathcal{B}$ , the only choice is by  $Nf_Y := F(f, \text{id}_Y) : F(X, Y) \rightarrow F(X', Y)$ . Which, given  $g : Y \rightarrow Y'$  in  $\mathcal{B}$ , the following diagram commutes using product category property:

$$\begin{array}{ccc} F(X, Y) & \xrightarrow{F(\text{id}_X, g)} & F(X, Y') \\ \downarrow F(f, \text{id}_Y) & & \downarrow F(f, \text{id}_{Y'}) \\ F(X', Y) & \xrightarrow{F(\text{id}_{X'}, g)} & F(X', Y') \end{array} \quad \equiv \quad \begin{array}{ccc} F_X(Y) & \xrightarrow{F_X g} & F_X(Y') \\ \downarrow Nf_Y & & \downarrow Nf_{Y'} \\ F_{X'}(Y) & \xrightarrow{F_{X'} g} & F_{X'}(Y') \end{array}$$

Hence,  $Nf : F_X \rightarrow F_{X'}$  is a natural transformation.

Then, we claim that  $N : \mathcal{A} \rightarrow \text{Cat}(\mathcal{B}, \mathcal{C})$  by  $N(X) := F_X$ ,  $Nf : F_X \rightarrow F_{X'}$  as above is a functor:

- $(N\text{id}_X)_Y = F(\text{id}_X, \text{id}_Y) = \text{id}_{F(X, Y)} = \text{id}_{F_X(Y)}$ , so  $N\text{id}_X = \text{id}_{F_X}$ .
- Given  $f : X \rightarrow X'$ ,  $f' : X' \rightarrow X''$ , then

$$N(f' \circ f)_Y = F(f' \circ f, \text{id}_Y) = F(f', \text{id}_Y) \circ F(f, \text{id}_Y) = Nf'_{Y'} \circ Nf_Y$$

So,  $N(f' \circ f) = Nf' \circ Nf$ .

Now, suppose two such functors  $F, F' : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$  has a natural transformation  $\mu : F \rightarrow F'$ , we want to construct a natural transformation  $\bar{\mu} : N \rightarrow N'$ , where  $N, N' : \mathcal{A} \rightarrow \text{Cat}(\mathcal{B}, \mathcal{C})$  are functors associated to  $F, F'$ . Which, given any  $X \in \mathcal{A}$ , we see the collection  $(\bar{\mu}_X)_Y := \mu_{(X, Y)}$  is a natural transformation  $F_X \rightarrow F'_X$ , due to the following diagram for all  $g : Y \rightarrow Y'$  in  $\mathcal{B}$ :

$$\begin{array}{ccc} F(X, Y) & \xrightarrow{F(\text{id}_X, g)} & F(X, Y') \\ \downarrow \mu_{(X, Y)} & & \downarrow \mu_{(X, Y')} \\ F'(X, Y) & \xrightarrow{F'(\text{id}_X, g)} & F'(X, Y') \end{array} \quad \equiv \quad \begin{array}{ccc} F_X(Y) & \xrightarrow{F_X g} & F_X(Y') \\ \downarrow (\bar{\mu}_X)_Y & & \downarrow (\bar{\mu}_X)_{Y'} \\ F'_X(Y) & \xrightarrow{F'_X g} & F'_X(Y') \end{array}$$

This shows  $\bar{\mu}_X : F_X \rightarrow F'_X$  is a natural transformation, or morphism within  $\text{Cat}(\mathcal{B}, \mathcal{C})$ . Then, notice that if  $\mu : F \rightarrow F'$ ,  $\mu' : F' \rightarrow F''$  are two natural transformations between functors  $F, F', F'' : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ , the following holds:

$$(\overline{\mu' \circ \mu})_Y = (\mu' \circ \mu)_{(X, Y)} = \mu'_{(X, Y)} \circ \mu_{(X, Y)} = (\bar{\mu'}_X)_Y \circ (\bar{\mu}_X)_Y$$

This shows as natural transformations,  $\bar{\mu}_X : N_X \rightarrow N'_X$  and  $\bar{\mu'}_X : N'_X \rightarrow N''_X$  satisfies  $(\overline{\mu' \circ \mu})_X = \bar{\mu'}_X \circ \bar{\mu}_X$ , so  $\overline{\mu' \circ \mu} = \bar{\mu'} \circ \bar{\mu}$  as natural transformations  $\bar{\mu} : N \rightarrow N'$ ,  $\bar{\mu'} : N' \rightarrow N''$ .

So, define  $\Phi(F : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}) := N : \mathcal{A} \rightarrow \text{Cat}(\mathcal{B}, \mathcal{C})$ , and  $\Phi(\mu : F \rightarrow F') := \bar{\mu} : N \rightarrow N'$  as above, it's a legitimate functor.

## II. Isomorphism on Objects:

Suppose  $\Phi(F) = \Phi(F')$ , then it's clear  $F, F'$  have the same action on the objects (as all  $X \in \mathcal{A}$  satisfies  $F_X = F'_X$ , so all  $Y \in \mathcal{B}$  satisfies  $F(X, Y) = F_X(Y) = F'_X(Y) = F'(X, Y)$ ). Now, given  $f : X \rightarrow X'$  in  $\mathcal{A}$  and  $g : Y \rightarrow Y'$  in  $\mathcal{B}$ , we also see that:

$$F(f, g) = F(f, \text{id}_{Y'}) \circ F(\text{id}_X, g) = Nf_{Y'} \circ F_X g = F' f_{Y'} \circ F'_X g = F'(f, \text{id}_{Y'}) \circ F'(\text{id}_X, g) = F'(f, g)$$

( $Nf = N'f$  is by  $\Phi(F) = \Phi(F')$ , then  $F_X = F'_X$  also the same logic).

This shows  $F = F'$ , as its map on morphisms are also the same.

Conversely, given

### III. Naturality:

If fix  $\mathcal{B}$  and varies  $\mathcal{A}, \mathcal{C}$ , since everything is constructed functorially, let's skip this part :)

### IV. Equivariance Version:

Assuming the first three statements, the remaining statement is to show:  $\Phi$  as a functor commutes with  $G$ -action.  $\square$

God the proof is actually long as hell...

**Definition 1.2.** Given a space  $X$ , the chaotic topological category  $\mathcal{E}X$  has object space  $X$ , morphism space  $X \times X$ , such that:

- Identity map  $i : X \rightarrow X \times X$  by  $x \mapsto (x, x)$  is continuous (also the same as diagonal morphism).
- The source map  $s : X \times X \rightarrow X$  by  $s(x, y) := x$  is continuous (also the same as first projection).
- The target map  $t : X \times X \rightarrow X$  by  $t(x, y) := y$  is continuous (also the same as second projection).
- Composition  $\circ : (X \times X) \times_{s,t} (X \times X) \rightarrow X \times X$  by  $\circ((x, y), (y, z)) := (x, z)$  is continuous.

Since there is only one morphism between any two objects, a morphism is equivalent to specify its source and target.

In particular, for any  $x \in X$ , the maps  $\eta_x, \mu_x : X \rightarrow X \times X$  by  $\eta_x(y) := (x, y)$ , and  $\mu_x(y) := (y, x)$  are continuous (as they're fixing one coordinate, while doing inclusion on the right). These also form natural isomorphisms  $\eta : x \rightarrow X$ , and  $\mu : X \rightarrow x$  as follow:

$$\begin{array}{ccc} x & \xrightarrow{(x,x)} & x \\ \eta_x(y)=(x,y) \downarrow & & \downarrow \eta_x(z)=(x,z) \\ y & \xrightarrow{(y,z)} & z \end{array}, \quad \begin{array}{ccc} y & \xrightarrow{(y,z)} & z \\ \mu_x(y)=(y,x) \downarrow & & \downarrow \mu_x(z)=(z,x) \\ x & \xrightarrow{(x,x)} & x \end{array}$$

This shows as a category,  $\mathcal{E}X$  is equivalent to the terminal category  $*$  (which only has one unique topological category structure).

Also, if  $X = G$  is a topological group (including discrete ones!) then  $\mathcal{E}G$  is isomorphic to the translation category of  $G$  (as all morphisms can be interpreted as a translation from one group element to the other).

Let  $\Pi$  be a topological group (intepreted as the category of single object  $*$ , with morphism space  $\Pi$ ), then there is a group homomorphism  $\Pi^{\text{op}} \rightarrow \text{Aut}_{\mathcal{T}\text{Cat}}(\mathcal{E}\Pi)$ , by right multiplication. So, all  $g \in \Pi$  satisfies  $g : \mathcal{E}\Pi \rightarrow \mathcal{E}\Pi$  by  $\sigma \mapsto \sigma g$  (which is a homeomorphism on  $\Pi$  by definition of topological group), and morphism  $(\sigma, \tau) \mapsto (\sigma g, \tau g)$  (again, this is the continuous diagonal action on  $\Pi \times \Pi = \text{Mor}(\mathcal{E}\Pi)$ ).

**Proposition 1.3.** *The orbit category  $\mathcal{E}\Pi/\Pi$  is isomorphic to  $\Pi$  as topological category.*

*Proof.* It's clear on the object space, as  $\text{Ob}(\mathcal{E}\Pi) = \Pi$  as set, so  $\text{Ob}(\mathcal{E}\Pi)\Pi = *$ .

The map on the morphism space is  $(\sigma, \tau) \sim (\sigma', \tau')$  iff there exists  $g \in \Pi$ , such that  $(\sigma, \tau) = (\sigma'g, \tau'g)$ . Which, this is equivalent to say  $\sigma\tau^{-1} = \sigma'gg^{-1}\tau'^{-1} = \sigma'\tau'^{-1}$ . So, we see all  $(\sigma, \tau) \sim (e, \tau\sigma^{-1})$ , and all  $g, g' \in \Pi$  satisfies  $(e, g) \sim (e, g')$  iff  $g^{-1} = g'^{-1}$ , or  $g = g'$ . So, we see each  $(e, g)$  establishes a distinct equivalent class, hence providing a bijection between  $(\Pi \times \Pi)/\Pi \cong \Pi$ , by  $(e, g) \mapsto g$ .

Finally, to show this is an isomorphism as topological category, we see such identification of morphism is topologically isomorphic (because  $\Pi \rightarrow \Pi \times \Pi \rightarrow \Pi$  by  $g \mapsto (e, g) \mapsto g$  is a homeomorphism). Then, given  $(e, g)$  and  $(e, g')$  in the quotient, since the second one is equivalent to  $(g, g'g)$ , then we see the composition is  $\circ((e, g), (g, g'g)) = (e, g'g)$ , which has quotient  $g'g$ , which is equivalent to the composition of the image of  $(e, g')$  and  $(e, g)$ , showing it's preserving the composition map. Hence, it's an isomorphism of categories.  $\square$

**Theorem 1.4.** *Given a  $G$ -space  $X$ , a topological group  $\Pi$  (regarded as a trivial  $G$ -space, by  $G \rightarrow e \rightarrow \text{Aut}(\Pi)$ ), the quotient functor  $p : \mathcal{E}\Pi \rightarrow \Pi$  induces an isomorphism if topological  $G$ -categories*

$$\text{Cat}(\mathcal{E}X, \mathcal{E}\Pi)\Pi \xrightarrow{\sim} \text{Cat}_G(\mathcal{E}X, \Pi)$$

*Proof.*

$\square$